Affordable Biosensing Technologies: Laser-Induced Graphene Platforms for Accessible Community Healthcare

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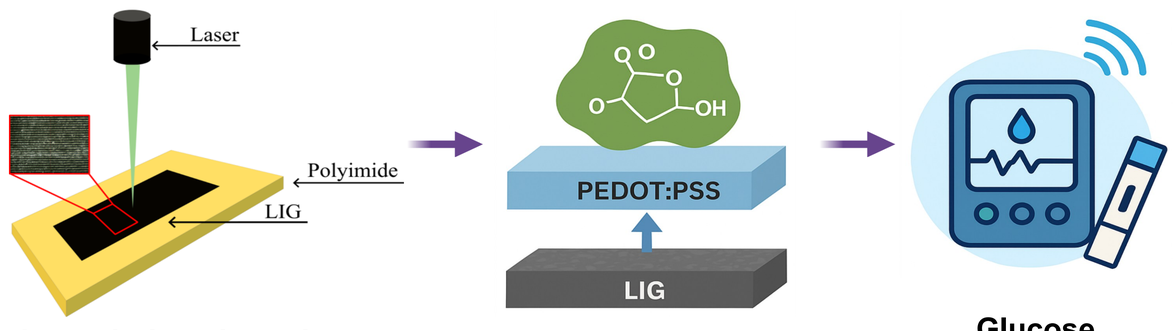
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1 Introduction

Access to healthcare remains a challenge in rural and underserved communities, where limited access to affordable diagnostic technologies can limit the timely detection and monitoring of chronic conditions such as diabetes. Addressing these disparities requires the development of low-cost, portable, and user-oriented biosensing platforms adapted to local healthcare needs. Laser-Induced Graphene (LIG) offers a promising approach to fabricating conductive, porous carbon-based materials through direct laser irradiation of polyimide (PI). Its electrical properties, tunable surface morphology, and compatibility with flexible substrates make it an attractive candidate for the development of electrochemical biosensing interfaces. Developing accessible diagnostic technologies involves more than creating effective materials. It is also important to understand the needs, preferences, and cultural context of the communities that may use them. Considering these factors can help guide the development of biosensing technologies that are not only affordable but also suitable for real-world community healthcare settings.

2 Objectives

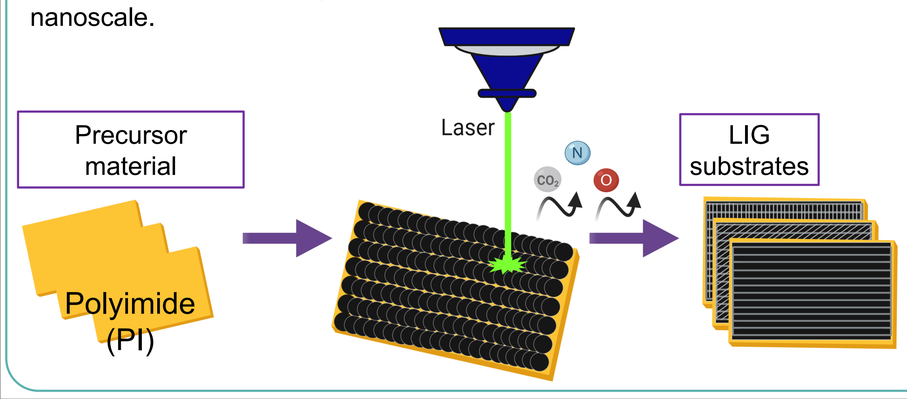
Technical: To synthesize, biofunctionalize, and characterize laser-induced graphene (LIG) substrates for non-invasive salivary glucose sensing.
Socio-Cultural: To evaluate user acceptability, interface readability, and socio-cultural health barriers through a community-based survey in indigenous Triqui populations (San Andrés Chicahuaxtla, Oaxaca).



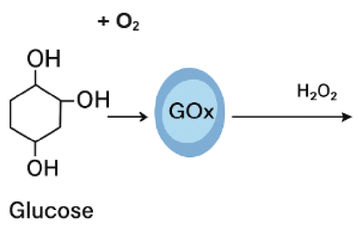
3 Materials and methods

LIG fabrication

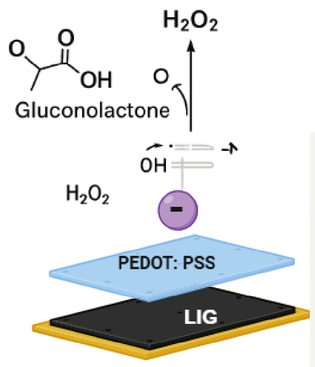
For the laser Induction process, we use a pulsed light laser, an instrument of precision that allows us to manipulate matter at the nanoscale.



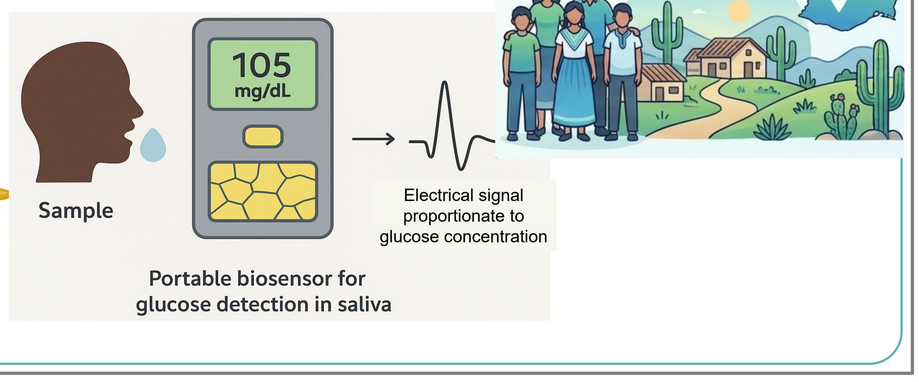
Surface functionalization



Glucose biosensor



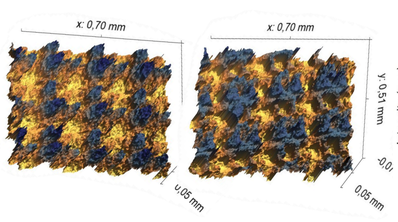
Community Survey



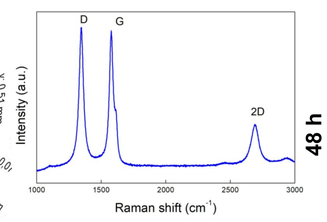
4 Results

1. Physicochemical characterization

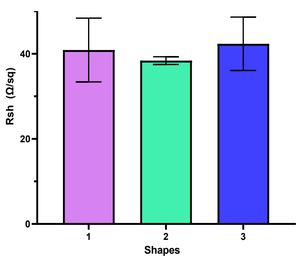
Roughness



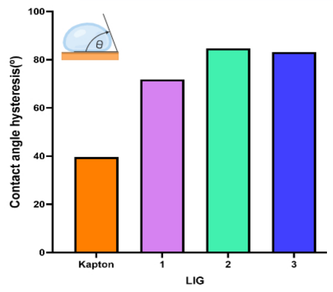
Raman spectroscopy



Conductivity

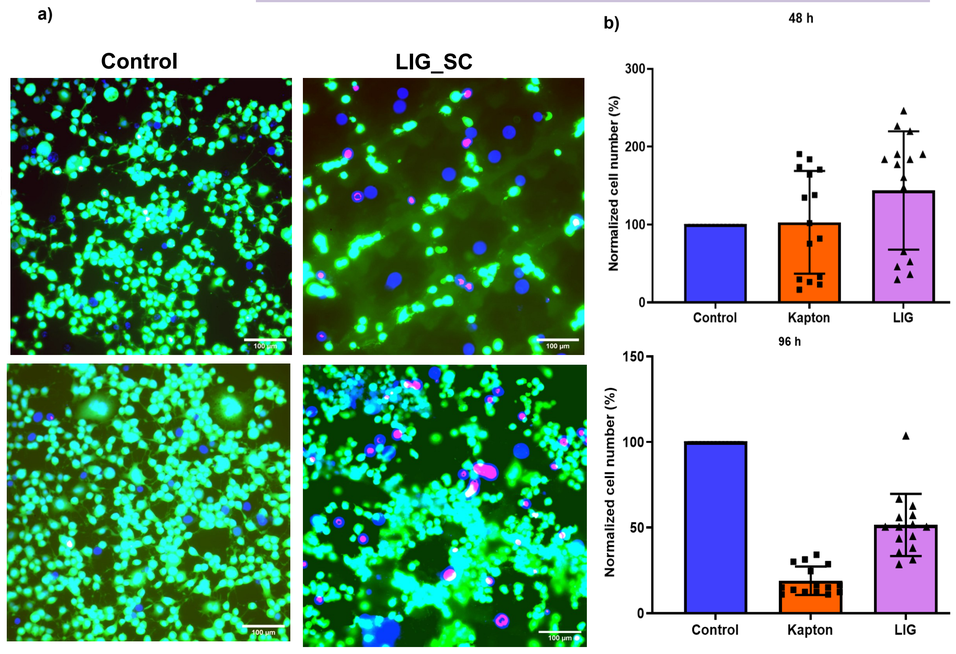


Wettability





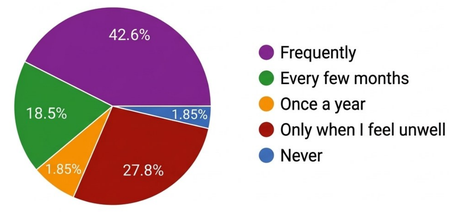
2. Biocompatibility of LIG substrates



a) Fluorescence microscopy images of N2a cells cultured on glass (control) and LIG substrates for 96 h. Hoechst (blue) stains cell nuclei, FDA (green) identifies metabolically active viable cells, and PI (red) labels dead cells. b–c) Quantitative analysis of cell density (% relative to control) after 48 h (b) and 96 h (c) of culture. ****p < 0.0001 vs. Control.

3. Community acceptability & socio-cultural survey

HOW OFTEN DO YOU (OR FAMILY) TEST FOR BLOOD SUGAR?
N=54 Community Participants

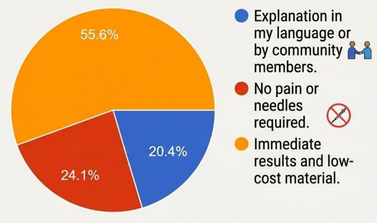


CURRENT TESTING HABITS (~61%)
Participants show established habits, with over 60% testing frequently or periodically.

OPPORTUNITY FOR PROACTIVE CARE
A non-invasive sensor could prevent complications by enabling regular monitoring before symptoms arise (27.8%).

COMMUNITY PERSPECTIVE ON TECH TRUST

What factor would give you the most confidence to use this new technology developed in laboratories?
N=54 Community Participants



KEY FINDING
(Summarized Results)
Primary Finding
Lower Cost & Immediate Results:
High community priority (55.6%)
Secondary Findings
Pain-Free Technology (24.1%)
Community Explanation (20.4%)
The community prioritizes affordability, quick feedback, and comfortable, locally explained technology.

Building trust requires immediate, low-cost, and comfort-focused design.

4 Conclusions

- LIG substrates were successfully fabricated by laser scribing of polyimide (Kapton®)
- Physicochemical characterization (Raman spectroscopy, XPS, and contact ang measurements) supported the formation of a porous, graphene-like conducti structure.
- Laser parameter optimization enabled control over electrical conductivity, surfac roughness, and wettability, demonstrating the versatility of LIG substrates.
- Preliminary biological assays indicated good biocompatibility with N2a cell supporting the potential of these substrates for biosensing applications.
- In a survey conducted in an Indigenous Triqui community (n = 54), responden prioritized rapid results and low cost (55.6%), painless, needle-free testing (24.1%), and communication in their own language or by community members (20.4%).
- These findings highlight the importance of integrating material innovation with affordability, usability, and cultural relevance to guide the development of more accessible biosensing technologies and support health equity.

5 References

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Acknowledgments

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